

LEDs used to fight infant jaundice

Many hospitals have begun using LEDs, instead of traditional fluorescent lamps, in phototherapy machines used to cure jaundice in newborn infants

New dengue vaccine to be tested

A highly effective new vaccine against dengue will be tested in India soon and may be available globally by 2015. Hopefully, this should help control the disease

Tomorrow's Tech

leg also features a built-in "anti-stumble" feature – if it senses that its user is starting to stumble, it lifts to avoid the obstruction and places the next step firmly to optimize balance. Studies, conducted by Nashville University, have shown that users equipped with the device walk 25 percent faster on level surfaces than when they're using traditional passive prosthetics. Research to improve the device continues at RIC, with bright hopes for easy future market availability.

Sensor that allows disease detection with the naked eye

With many diseases, the quicker the diagnosis, the better it can be treated. Unfortunately, a lot of diseases, and especially AIDS, are notoriously hard to diagnose in their initial stages. Therefore, usually, by the time the disease is detected via visible physical symptoms, the patient may just be beyond help. In this fight against time, British scientists have come up with a solution that might just give early diagnosis the slight edge it needs.

Created at Imperial College, London, the super-sensitive prototype sensor lets doctors detect early stage diseases with the naked eye, using nanotechnology to analyze blood samples. A positive reaction to p24, a protein that indicates early HIV infection, or PSA, a protein that can indicate prostate cancer, generates irregular clumps of nanoparticles that emit a blue colour in a solution kept in a disposable container. A negative reaction, however, causes the nanoparticles to separate into ball-like shapes, creating



The colour coding makes disease detection quick and simple for the naked eye - red is safe, and blue is 'Yeah, you're infected...'

a reddish hue. The colours are easily distinguishable at a glance.

Simple and quick HIV tests using saliva already exist; however, they can only detect the virus if it is present in highly concentrated amounts. Concentrations of p24 are high in the first few weeks after infection, so a test like this one could be used for quick diagnosis at early stages, without having to wait for the virus to spread in the patient's body.

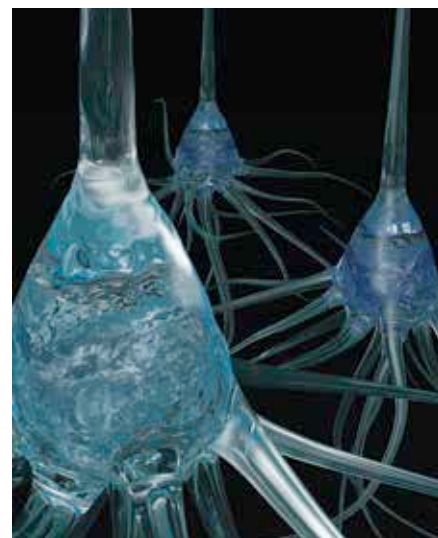
The team behind the project says that their technology is 10 times more sensitive than current methods used for diagnosis via similar biomarking systems, and in fact showed viral loads in HIV-infected patients that were previously undetectable by the far more popular enzyme-linked immunosorbent assay (ELISA) test, as well as another gold-standard nucleic acid-based test. Hopeful news indeed...

New electrode may help speed of transfer of brain signals

Researchers at the University of Michigan have created a new thin, flexible electrode that they say is 10 times smaller than the nearest competing device, and is likely to finally make long-term mapping of neural activity a practical possibility. This technology could eventually be used to help speed up the rate at which brain signals are communicated to external devices, like prosthetic limbs, which would enable a new generation of artificial devices, like the bionic leg mentioned previously, to be developed.

The problem with traditional electrodes is one of size and incompatibility with neurons. They are not only enormous in comparison, but are also stiff and can cause inflammation in the surrounding region, blocking communication between the electrode and the surrounding cells.

The new electrode, in comparison, is extremely unobtrusive and accommodating. It is made of a thread of highly conductive carbon fibre, coated in plastic to block out all other interfering signals. The conductive gel pad at the end enables a close connection between the electrode and the surrounding cell membranes, resulting in clearer and quicker transmis-



An artist's rendition of individual neurons – the new electrode will make communication between them a lot faster

sions between brain cells. The technology has been tested on rats, with stunning results. In tests, the narrow profile of the electrode allowed it to focus on just one neuron, resulting in sharp and clear electrical signals that could be easily measured and analyzed.

Apart from enabling more advanced methods of control for prosthetics, this technology also has the potential to add greatly to our knowledge of the workings of the human brain. Analysis of the data generated from these electrodes can solve problems whose solutions have long eluded the medical community, such as how do neurons communicate with one another, and what "pathways" are used to relay information across the brain? It appears that we may finally get our answers.

What the future holds...

These are just some of the myriad new discoveries and inventions occurring in the medical field today. As we progress, we keep looking for new ways to look and feel younger, be more energetic and youthful, and live longer (maybe forever). For all the progress we have made, there still exist entire realms of exciting opportunity and untold possibility with new and developing fields like nanotechnology, genetics and biotechnology. We have barely scratched the surface here, and the journey is just beginning. 